

SHREYA SHUKLA

Theoretical physics | Quantum learning

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RESEARCH INTERESTS

Quantum Hamiltonian learning and Quantum Simulations; Quantum error correction; inverse problems in statistical and quantum field theory; renormalization-group flows and effective field theories; lattice gauge theory; machine learning for interacting physical systems.

ACADEMIC APPOINTMENTS

University of Arizona, Department of Mathematics 2026–Present

Research Scientist. Research on Quantum and Classical Algorithms and Optimization Problems.

Los Alamos National Laboratory, Theoretical Division (T-5) 2024–2026

Postdoctoral Researcher. Research on classical and quantum Hamiltonian learning, statistical field theories, and inference from correlations.

University of California, Irvine, Department of Physics 2019–2024

Teaching Assistant and Graduate Researcher. Research on string model building and supersymmetric theories and non-perturbative physics.

EDUCATION

Ph.D., Theoretical Particle Physics, University of California, Irvine June 2024

MS, Mathematical Physics, University of Edinburgh September 2018

BS, Physics, Indian Institute of Technology, Kanpur June 2017

PREPRINTS AND MANUSCRIPTS

- Scalable Learning of Quantum Hamiltonians via Operator Score Matching.**
S. Shukla, A. Jayakumar, and A. Y. Lokhov.
Submitted to *Nature Physics*. arXiv:2609.25246.
- Learning of Statistical Field Theories.**
S. Shukla, A. Jayakumar, and A. Y. Lokhov.
Revised manuscript under review at *Nature Communications*. arXiv:2511.09859.

PEER-REVIEWED PUBLICATIONS

- Computationally Sufficient Statistics for Ising Models.**
A. Jayakumar, **S. Shukla**, M. Vuffray, A. Y. Lokhov, and S. Misra.
Accepted at *NeurIPS 2026* (Poster). arXiv:2602.12449 (2026).
- Efficient Learning of Lattice Gauge Theories with Fermions.**
S. Shukla, Y. Yamauchi, A. Y. Lokhov, S. Lawrence, and A. Jayakumar.
Published in *Physical Review D* **114**, 014514 (2026). arXiv:2512.19891 (2025).
- Chirality Changing RG Flows: Dynamics and Models.**
Y. Shirman, **S. Shukla**, and M. Waterbury.
JHEP **06** (2023), 168. arXiv:2303.08847.
- Neutrino Flavor Model Building and the Origins of Flavor and CP Violation.**
Y. Almumin, M.-C. Chen, M. Cheng, V. Knapp-Perez, Y. Li, A. Mondol, S. Ramos-Sanchez, M. Ratz, and **S. Shukla**.
Universe **9**, 512 (2023). arXiv:2204.08668.
- Quasi-Eclectic Modular Flavor Symmetries.**
M.-C. Chen, V. Knapp-Perez, M. Ramos-Hamud, S. Ramos-Sanchez, M. Ratz, and **S. Shukla**.
Physics Letters B **824**, 136843 (2022). arXiv:2108.02240.

6. Generation Flow in Field Theory and Strings.

S. Ramos-Sanchez, M. Ratz, Y. Shirman, **S. Shukla**, and M. Waterbury.
JHEP **10** (2021), 144. arXiv:2109.01681.

7. Metaplectic Flavor Symmetries from Magnetized Tori.

Y. Almumin, M.-C. Chen, V. Knapp-Perez, S. Ramos-Sanchez, M. Ratz, and **S. Shukla**.
JHEP **05** (2021), 078. arXiv:2102.11286.

SELECTED RESEARCH CONTRIBUTIONS

Quantum Hamiltonian Learning via Operator Score Matching

2026

- Developed an operator score-matching framework to infer Hamiltonian parameters from Gibbs-state data without evaluating the partition function.
- Derived identifiability conditions and local convergence guarantees near the true parameters; analyzed how the choice of operator directions affects parameter recovery.
- Constructed approximations based on nested commutators and polynomials, using locality to control the support of the required observables under appropriate assumptions.
- Benchmarked parameter recovery in transverse-field Ising, XXZ, Fermi–Hubbard, and lattice ϕ^4 models, including matrix product state simulations and thermal sampling.

Learning Statistical Field Theories and Renormalization-Group Flows

2025–2026

- Developed and implemented score-matching and pseudolikelihood methods for learning couplings in discrete, continuous, and mixed field theories without computing partition functions.
- Expressed learning objectives in terms of observable correlations, enabling inference without access to full configurations and achieving errors below 1% in benchmark models.
- Combined Monte Carlo sampling, optimization, and real-space coarse-graining to reconstruct nonperturbative RG flows, phase boundaries, and emergent interactions.
- Studied Ising gauge, ϕ^4 , Schwinger, Sine–Gordon, and mixed spin–gauge models.

Learning Lattice Gauge Theories with Fermions

2025–2026

- Constructed convex learning objectives from Schwinger–Dyson relations and established score matching as a special case of a broader family of estimators.
- Extended the framework to Grassmann-valued fields and gauge theories, including formulations for lattice QCD; demonstrated parameter recovery in the lattice Thirring model.

Computationally Sufficient Statistics for Ising Models

2026

- Studied the trade-off between computational and observational resources for learning Ising models from limited statistics.
- Established parameter recovery using statistics up to order $O(\gamma)$ for models of ℓ_1 width γ .
- Examined how prior structural information reduces the observational requirements for learning.

ONGOING RESEARCH

Quantum Control and Flow Matching. Using flow matching and/or score matching to steer an initial density matrix to a target density matrix using Hamiltonian controls in open systems.

Glueball Effective Field Theory from Lattice Yang–Mills. Developing coarse-graining maps from gauge configurations to effective glueball fields using smeared, symmetry-preserving interpolating operators. Applying score matching to infer masses and interactions, with numerical implementation in Julia.

Inverse Scattering and S-Matrix Learning. Investigating reconstruction of effective interaction potentials from scattering observables and boundary measurements using Green’s-function identities and inverse PDE methods.

COMPUTATIONAL METHODS AND SOFTWARE

Languages: Julia, Python, Mathematica, C.

Methods: Monte Carlo and Hamiltonian Monte Carlo; matrix product states and METTS sampling; score matching and pseudolikelihood; numerical optimization; real-space RG; symbolic calculations.

Software: NumPy, SciPy, Matplotlib, Pandas, JuMP, scikit-learn, PyTorch, TensorFlow, Keras.

CERTIFICATIONS AND TRAINING

Erdős Institute Quantum Computing Bootcamp	In progress
Erdős Institute Deep Learning Bootcamp	2024
Erdős Institute Data Science Bootcamp	2023

EARLIER RESEARCH

Chirality, Strong Dynamics, and Flavor Symmetries, UC Irvine 2019–2024

Supervisors: Michael Ratz and Yuri Shirman. Studied chirality-changing RG flows and the stability of chirally symmetric vacua in supersymmetric theories. Constructed string models exhibiting generation flow, derived Yukawa couplings on magnetized tori, and investigated metaplectic and quasi-eclectic flavor symmetries.

Fundamentals of Conformal Invariance, University of Edinburgh 2018

Supervisor: Luigi Del Debbio. Studied anomalous Ward identities, spontaneous symmetry breaking, and the dependence of current correlators on Nambu–Goldstone boson and dilaton masses.

Turbulence in Burgers’ Equation, ICTS Bangalore Summer 2016

Supervisor: Samriddhi Sankar Ray. Analyzed velocity gradients and shock formation in one- and higher-dimensional Burgers’ equations.

Acoustic Analogue Spacetimes, Dissipative Nonlinear Dynamical Systems, IIT Kanpur 2015–2016

Supervisor: Tapobrata Sarkar and Sagar Chakraborty. Studied acoustic perturbations, frame dragging, ergospheres, and horizons in flowing fluids. Also investigated Hannay’s angle, Berry curvature, and geometric effects in adiabatic evolution and dissipative systems.

AWARDS AND FELLOWSHIPS

Dissertation-year fellowship support, UC Irvine	2023–2024
Honorable Mention for Outstanding Teaching, UC Irvine	2023
Outstanding Graduate Student Award, UC Irvine	2020
Chancellor’s Fellowship, UC Irvine	2019
DST INSPIRE Fellowship	2013

TEACHING AND MENTORING

University of California, Irvine. Led discussion sections, held office hours, and supported students in Statistical Mechanics (Physics 214A, 2021), Mathematical Physics (125A, 2020), Quantum Mechanics (113A, 2020), and Classical Mechanics (112A, 2020). Mentored undergraduates interested in physics graduate study, including through Women in Physics and Astronomy (WiPA). Organized WiPA outreach and social events at UCI; served as student head of the physics DUGC and organized Astronomy Club workshops and lectures at IIT Kanpur.

TALKS AND PRESENTATIONS

Learning Statistical Field Theories, APS March Meeting	2025
Chirality Changing RG Flows, TASI	2023
Metaplectic Flavor Symmetries in Magnetized Tori, Amplitudes Summer School, Prague	2022
Misaligned Supersymmetry in String Theory, UC Irvine	2021
Yukawa Couplings from Magnetized Tori, APS DPF Meeting	2021
Generation Flow in Field Theories and Strings, Cargèse Summer School	2021
Metaplectic Flavor Symmetries, UNAM, Mexico	2021

SELECTED WORKSHOPS AND SCHOOLS

APS March Meeting (2025); Physics Informed Machine Learning, LANL (2024); PiTP: Understanding Confinement, Princeton (2023); TASI: Aspects of Symmetry (2023); Amplitudes Summer School, Prague (2022); KITP: Confinement, Flux Tubes, and Large N (2022); Cargèse: New Frontiers in Quantum Field Theory (2021).